Lecture 7: Time Complexity, P vs. NP

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Time Complexity

Computability: worst case

Number of configurations TM needs to reach an accept/reject state on this input

Recognizers for a language L;
Which one is better?

	M_A	M_B
ε	$< \infty$	$< \infty$
0	$< \infty$	$< \infty$
1	$< \infty$	∞
00	$< \infty$	$< \infty$
01	$< \infty$	$< \infty$
10	$< \infty$	$< \infty$
11	$< \infty$	$< \infty$
000	$< \infty$	∞
001	$< \infty$	$< \infty$
010	$< \infty$	$< \infty$
$\vdots$	$\vdots$	$\vdots$

Does not halt on this input and, hence, is not a decider

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11	$< \infty$	$< \infty$
000	$< \infty$	∞
001	$< \infty$	$< \infty$
010	$< \infty$	$< \infty$
$\vdots$	$\vdots$	$\vdots$

Does not halt on this input and, hence, is not a decider

In the rest of the course material we just consider decidable languages

A decidable language L

Number of configurations TM needs to reach an accept/reject state on this input

Deciders for L:

Inputs	M_A	M_B	M_C	M_D	M_E	M_F	M_G	M_H	
ε	2	2	5	2	3	4	2	2	⋯
0	2	5	12	2	3	5	12	5	⋯
1	20	12	14	13	8	19	2	9	⋯
00	32	14	18	9	18	3	5	90	⋯
01	12	21	56	8	12	18	18	30	⋯
10	21	22	26	15	11	12	32	15	⋯
11	11	12	25	100	13	48	98	29	⋯
000	320	201	159	201	190	200	180	65	⋯
001	211	208	190	200	189	301	219	82	⋯
010	328	271	214	441	193	208	109	77	⋯
011	227	261	191	201	188	107	211	207	⋯
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

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1	20	12	14	13	8	19	2	9	⋯
00	32	14	18	9	18	3	5	90	⋯
01	12	21	56	8	12	18	18	30	⋯
10	21	22	26	15	11	12	32	15	⋯
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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

How to compare different deciders?

Two deciders for L

	M_A	
ε	2	2
0	4	12
1	12	
00	15	110
01	22	
10	17	
11	110	
000	49	300
001	34	
010	38	
011	300	
⋮		
	⋮	

	M_B	
ε	2	2
0	9	10
1	10	
00	40	45
01	30	
10	45	
11	39	
000	73	85
001	77	
010	85	
011	80	
⋮		
	⋮	

Worst case in this group

Running time of a TM

Definition: Let M be a TM that halts on all inputs (decider). The running time or time complexity of M is the function $t : \mathbb{N} \to \mathbb{N}$ where

$$t(n) = \max_{w \in \Sigma^* : |w| = n} \text{number of steps } M \text{ takes on } w$$

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- M runs in time $t(n)$
- n represents the input length

$$t(0) = 2$$
$$t(1) = 10$$
$$t(2) = 45$$
$$t(3) = 85$$

ε	2	2
0	9	10
1	10	
00	21	45
01	30	
10	45	
11	33	
000	73	85
001	77	
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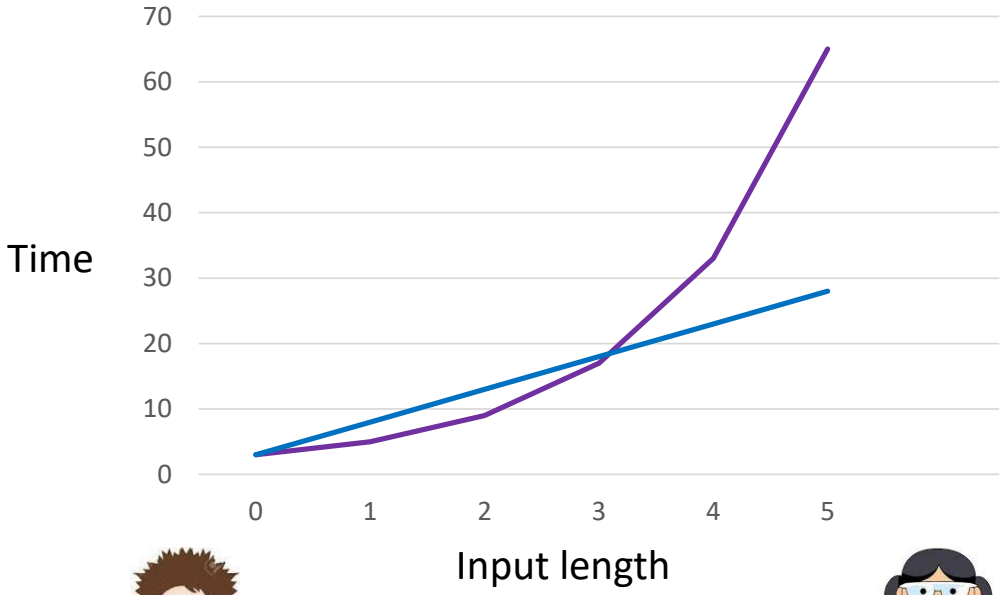
$\vdots$		

Two deciders for L

M_A	
Length 0	$t_1(0) = 3$
Length 1	$t_1(1) = 5$
Length 2	$t_1(2) = 9$
Length 3	$t_1(3) = 17$
Length 4	$t_1(4) = 33$
Length 5	$t_1(5) = 65$
$\vdots$	$\vdots$

M_B	
Length 0	$t_2(0) = 3$
Length 1	$t_2(1) = 8$
Length 2	$t_2(2) = 13$
Length 3	$t_2(3) = 18$
Length 4	$t_2(4) = 23$
Length 5	$t_2(5) = 28$
$\vdots$	$\vdots$

$t_1(n) = 2^{n+1} + 1$ vs $t_2(n) = 5n + 3$



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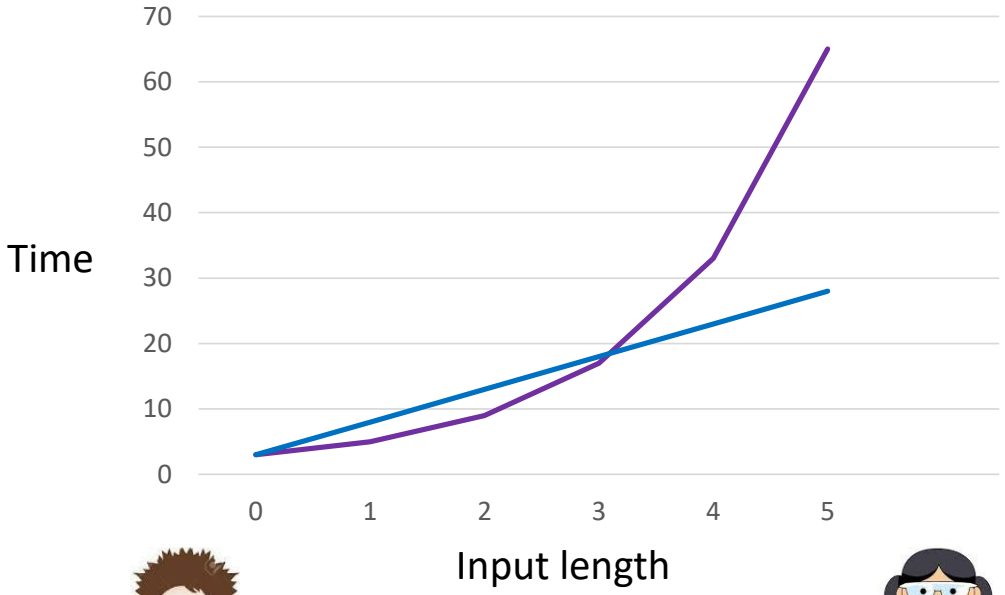

How to compare running time functions?

The Growth of Functions

- Three TMs: A, B, C with different running times in ms (say 1 step is 1 ms)

	A ($1000 \log_2 n$ ms)	B ($2n^2$ ms)	C (2^n ms)

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	A ($1000 \log_2 n$ ms)	B ($2n^2$ ms)	C (2^n ms)
$n = 2$	1 s	8 ms	4 ms
$n = 4$	2 s	32 ms	16 ms
$n = 8$	3 s	128 ms	256 ms
$n = 16$	4 s	512 ms	1 m 5s 536 ms

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$n = 64$	6 s	8 s 192 ms	$>$ age of the universe

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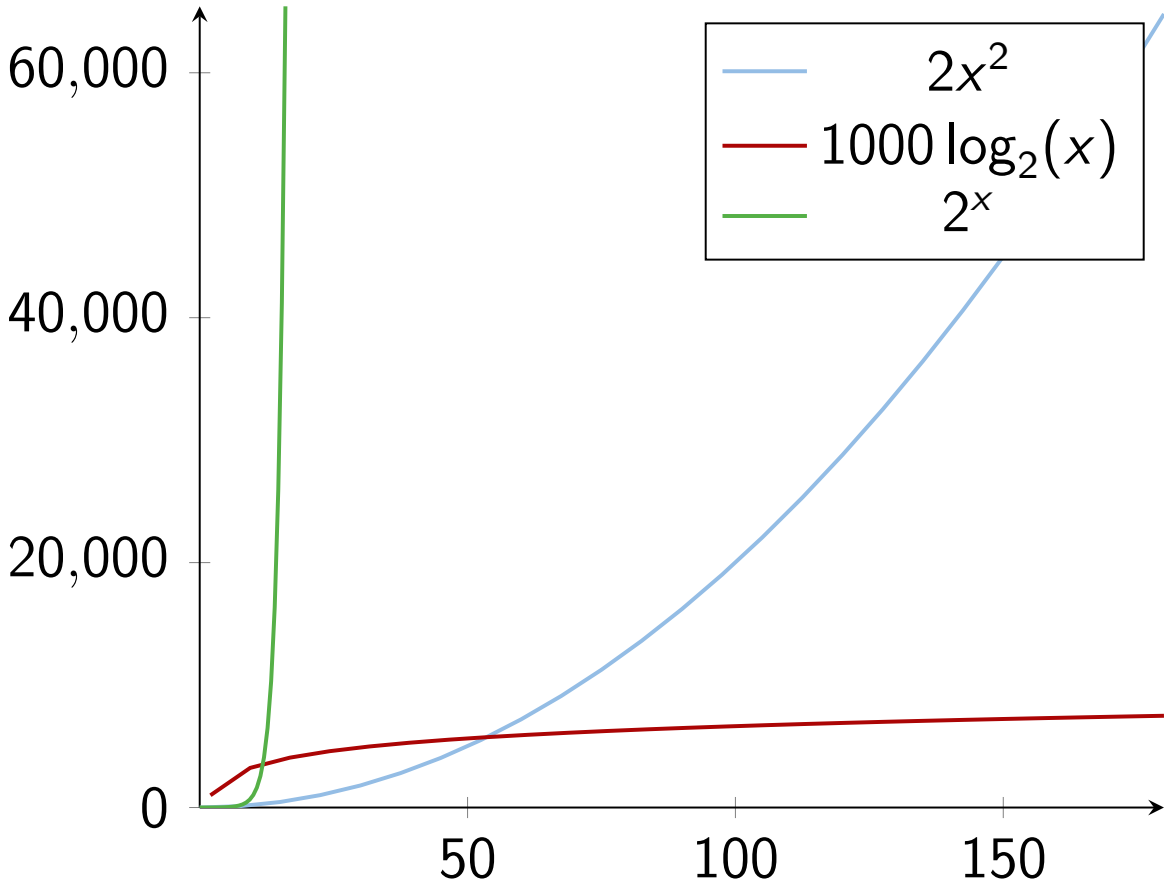

Big-O and Small-o notation

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Definition (Big-O): Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$. We say $f(n) = O(g(n))$ if

$$\exists C > 0, n_0 \in \mathbb{N} \text{ s.t. } \qquad \forall n \geq n_0 \;\; f(n) \leq C \cdot g(n)$$

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Examples:

- $5n^3 + 1 = O(2^n)$? YES
- $5n^3 + 1 = O(20n + 5)$? NO

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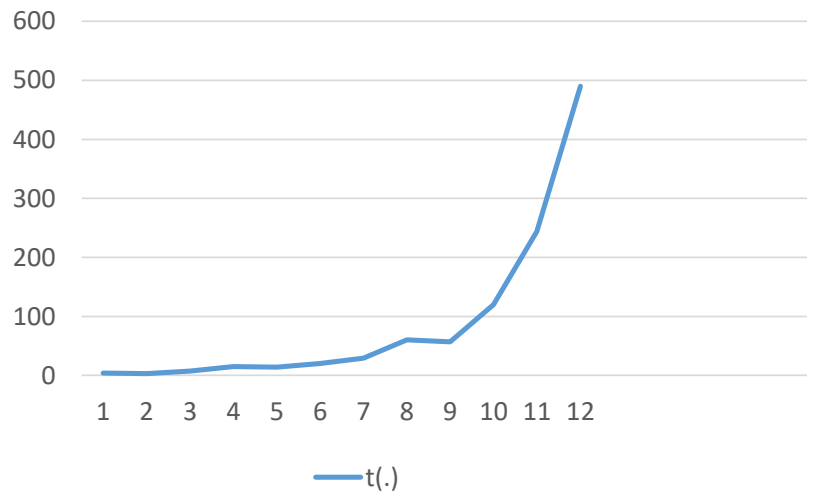
Examples:

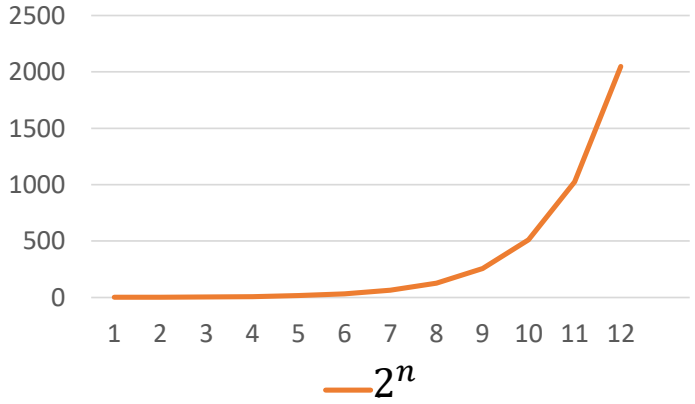
- $\sqrt{n} = o(n)$? YES
- $f(n) = o(f(n))$? NO

To summarize. . .

M		
2	Length 0	$t(0) = 2$
4 3	Length 1	$t(1) = 4$
2 3 ⋮	Length 2	$t(2) = 3$
4 ⋮ 7 ⋮	Length 3	$t(3) = 7$
6 10 ⋮ 15 ⋮	Length 4	$t(4) = 15$
10 8 ⋮ 14 ⋮	Length 5	$t(5) = 14$ ⋮
	⋮	



$t(n) = O(2^n)$



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Definition: Time complexity class

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Example:

$TIME(n) \subseteq TIME(n^2) \subseteq \cdots \subseteq TIME(2^{\sqrt{n}}) \subseteq TIME(2^n) \subseteq TIME(2^{2^n}) \ldots$

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Example: $REGULAR \subseteq TIME(n)$

The complexity class **P** and efficiency

Definition: **P** is the class of languages that are decidable in polynomial time on a (deterministic) Turing machine. In other words,

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Some languages in P:

- $\{\langle A \rangle \ : \ A \text{ is a sorted array of integers}\}$
- $\{\langle G, s, t \rangle \ : \ s \text{ and } t \text{ are vertices connected in graph } G\}$ (Breadth-First Search)
- $\{\langle G \rangle \ : \ G \text{ is a connected graph}\}$

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Robust definition: **P** is invariant for all models of computation that are polynomially equivalent to deterministic TM

To show $L \in \mathbf{P}$ enough to describe deterministic algorithm in words and in your favorite programming/pseudo-code language.

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Extended Church–Turing thesis: **P** corresponds to the class of problems that are "realistically" solvable in our universe

Possible challenges: randomisation and quantum?

NP: Verification vs. Search

SAT-verify and SAT

Conjunctive Normal Form (CNF) Formula:

$$\varphi_1 = (\overline{x} \vee \overline{y} \vee z_0) \wedge (x \vee \overline{y} \vee z_1) \wedge (\overline{x} \vee y \vee z_2) \wedge (x \vee y \vee z_3)$$

$$\varphi_2 = \overline{x_1} \wedge (x_1 \vee \overline{x_2}) \wedge (x_1 \vee x_2 \vee \overline{x_3}) \wedge (x_1 \vee x_2 \vee x_3 \vee \overline{x_4})$$

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Check: φ_1 has 32 satisfying assignments, φ_2 has only one, φ_3 has zero.

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A formula is **satisfiable** if it has at least one satisfying assignment.

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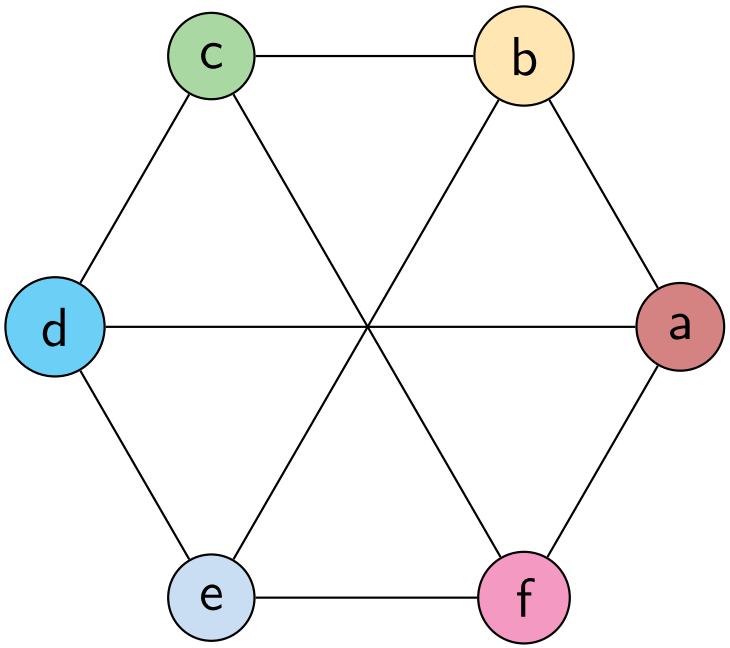
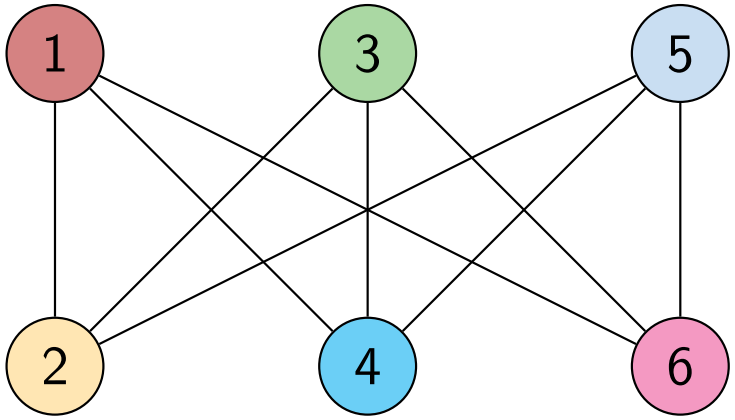
Is SAT in P?

Decider for SAT:

1. For each assignment C:
   - If $\langle \varphi, C \rangle \in$ SAT-verify, ACCEPT φ.
2. REJECT φ.

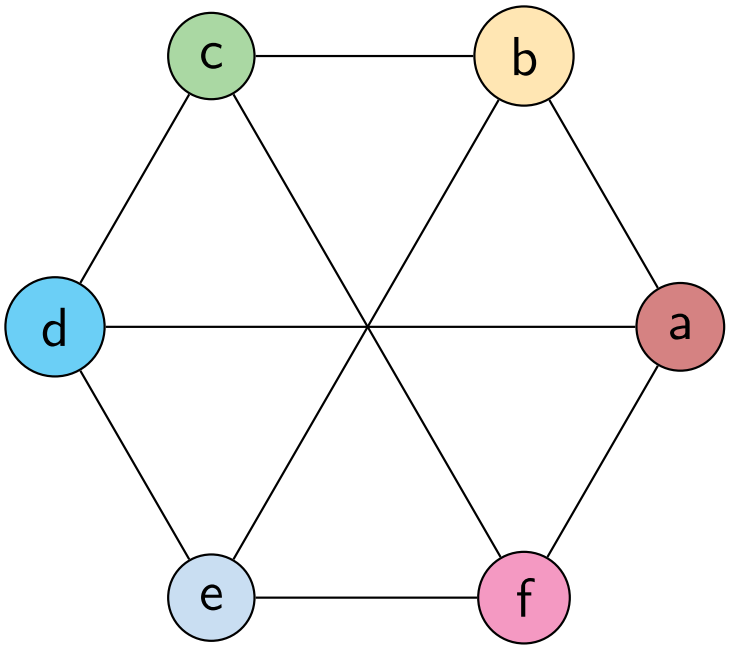
GI-verify and GI

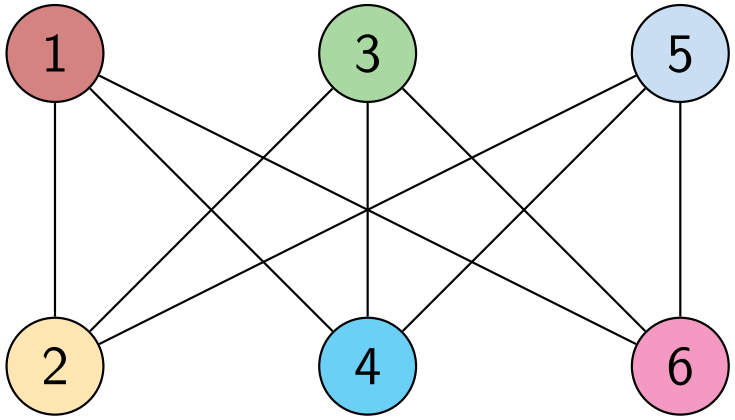




Graph Isomorphism: Bijection $f : V(G_1) \longrightarrow V(G_2)$ which preserves adjacency: $\{u, v\} \in E(G_1) \Leftrightarrow \{f(u), f(v)\} \in E(G_2)$

GI-verify and GI





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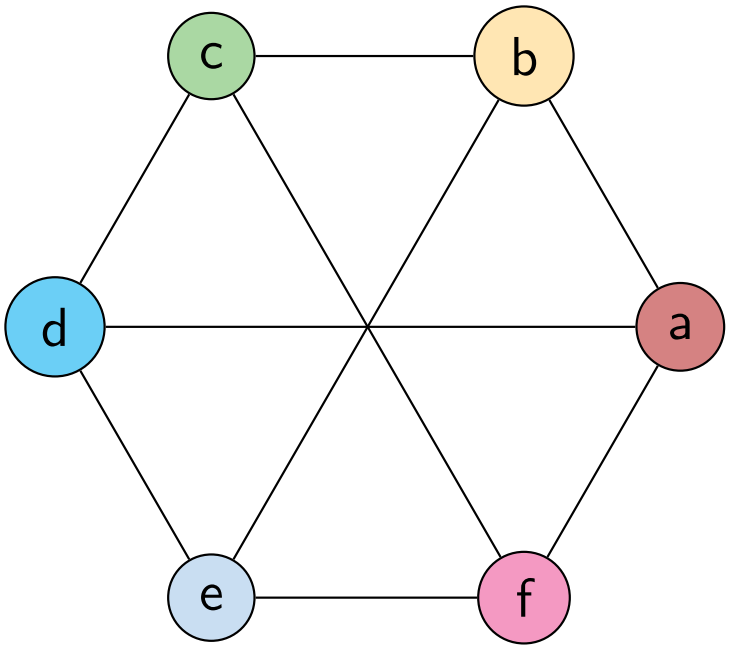
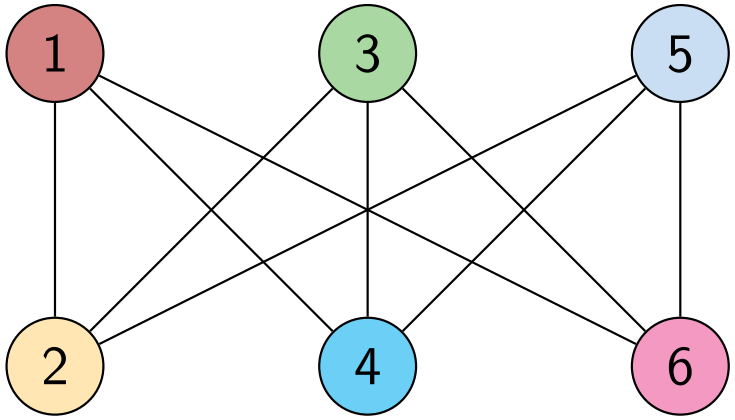
Eg. $a \to 1$ $b \to 2$ $c \to 3$ $d \to 4$ $e \to 5$ $f \to 6$ in the graphs above.

GI-verify and GI





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Two graphs are **isomorphic** if they have at least one graph isomorphism.

GI-verify and GI

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Is GI-verify in P? Yes!

1. Check that C is a bijection: For each $u, v \in V(G_1)$:
   - Check $\{u, v\} \in E(G_1) \Leftrightarrow \{C(u), C(v)\} \in E(G_2)$.

GI-verify and GI

$$\text{GI-verify} = \{\langle G_1, G_2, C\rangle \ : \ C : V(G_1) \longrightarrow V(G_2) \text{ is a graph isomorphism}\}$$

Is GI-verify in P? Yes!

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Decider for GI:

1. For each function $C : V(G_1) \longrightarrow V(G_2)$:
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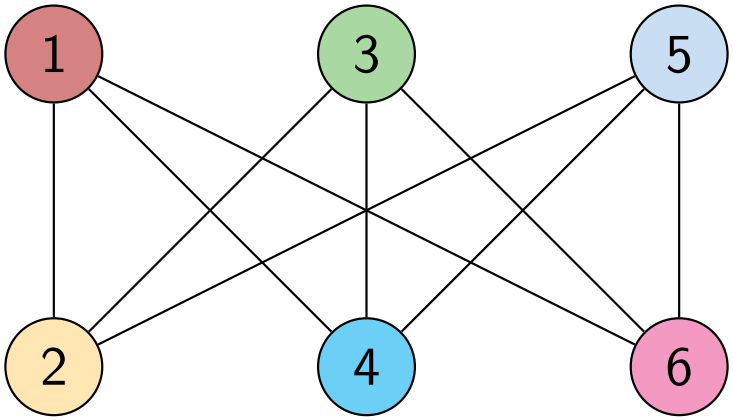
2. REJECT $\langle G_1, G_2\rangle$.

INDSET-verify and INDSET



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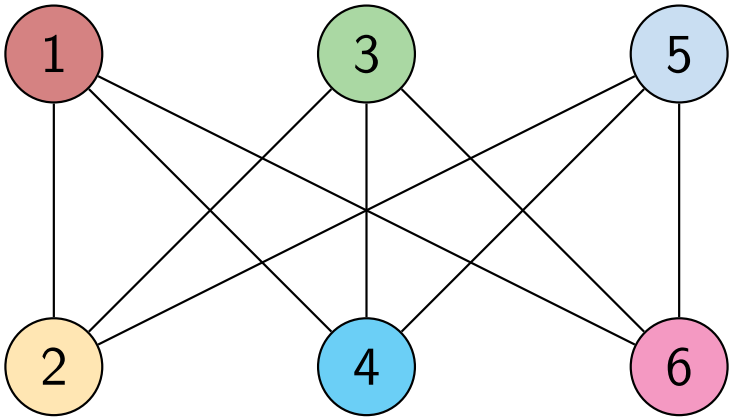

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Eg. $\{1, 3, 5\}$, $\{2, 4\}$, $\{6\}$, $\emptyset$, etc. in the graph above.

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Verifiers and the class NP

Recall: A **decider** for language L is a TM M such that for each $x \in \Sigma^*$

- If $x \in L$, then M accepts x.
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Why is it called **NP**?

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M accepts $\langle x, C \rangle$ iff $x \in L$ and C is an accepting path of N on x.

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$\therefore$ SAT, GI, INDSET are all in NP.

P and NP

Is $P \subseteq NP$? Yes (obviously)

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Is $\mathsf{P} = \mathsf{NP}$? Nobody knows ...

Find the answer and win USD 1,000,000!

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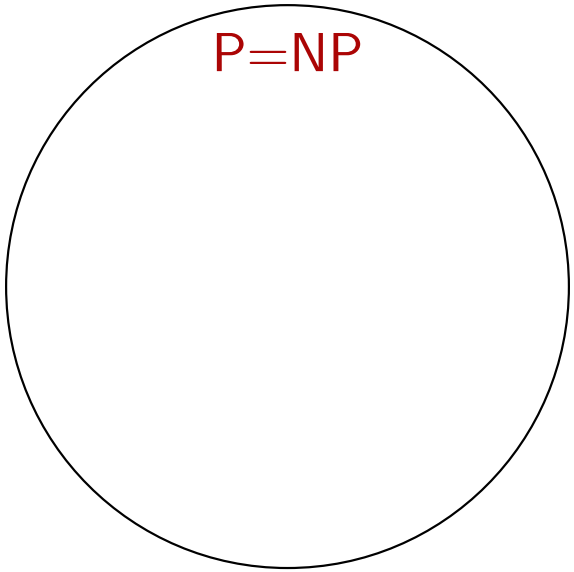

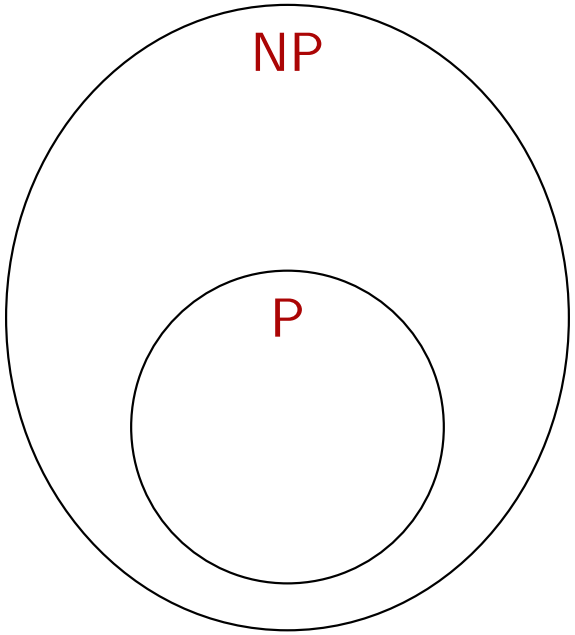

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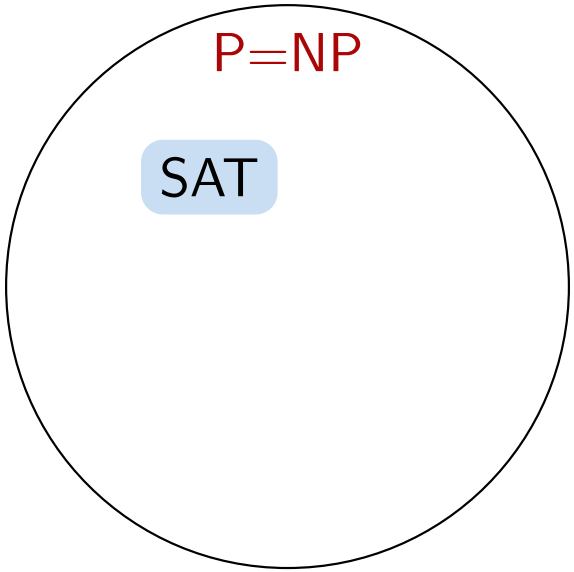

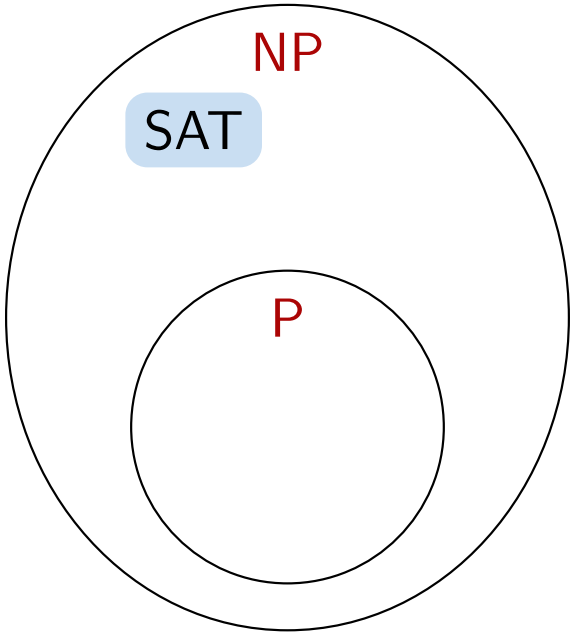

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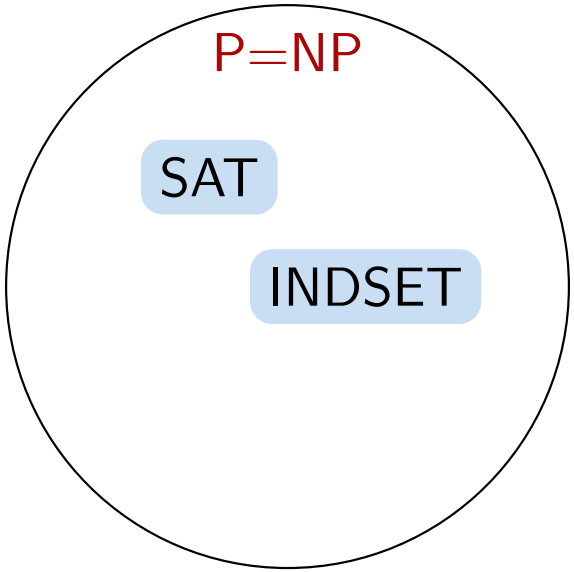

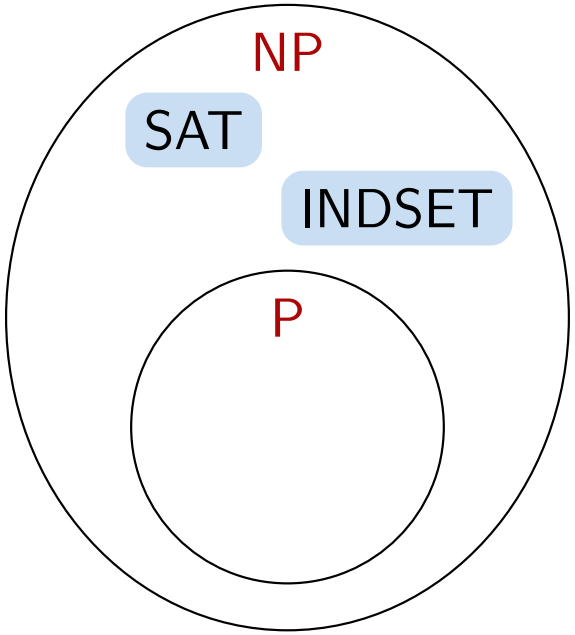

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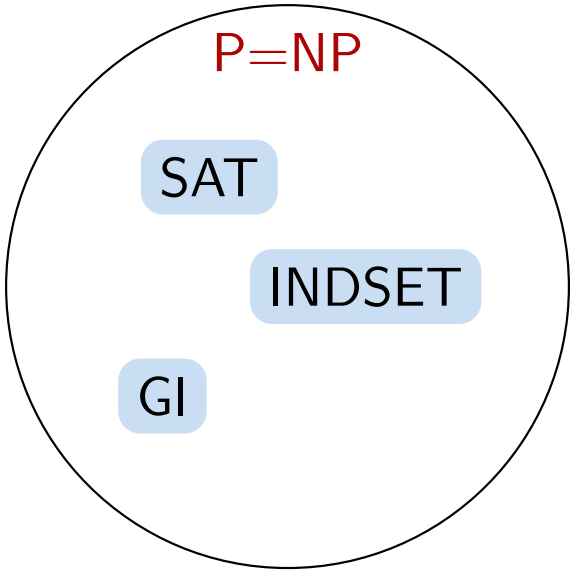

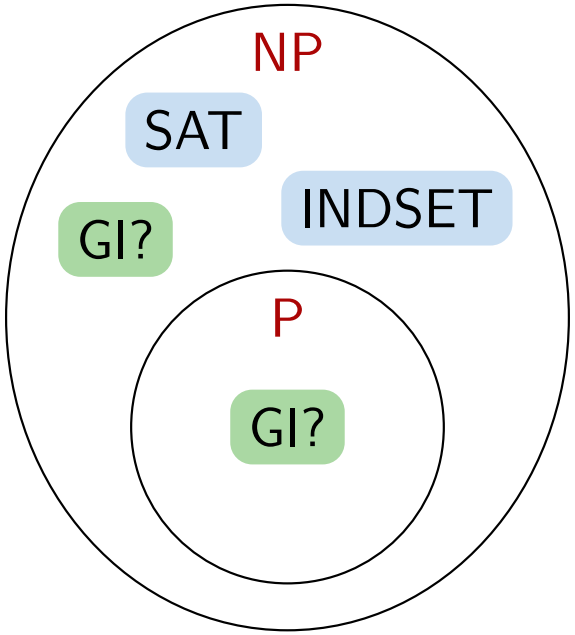

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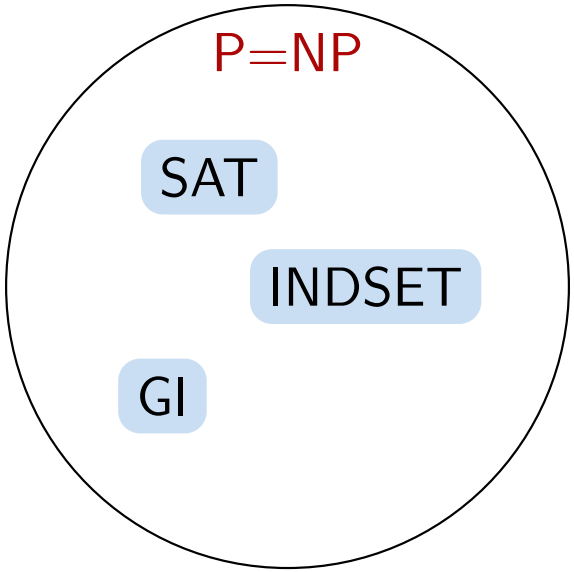

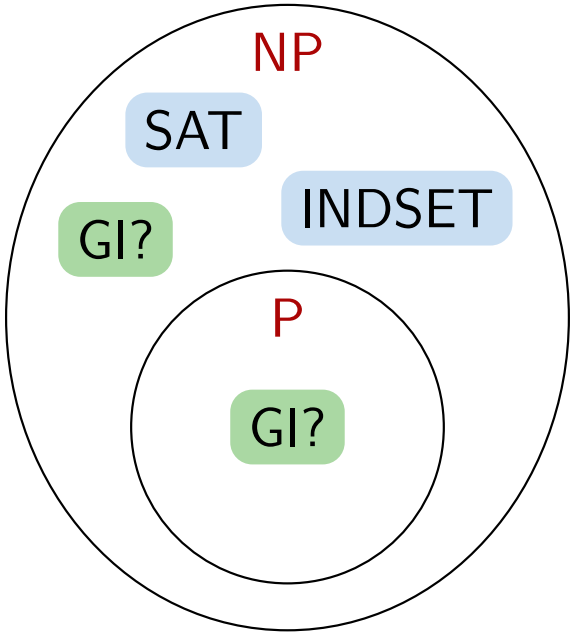

Cook-Levin Theorem (informal): $SAT \in P$ iff $P = NP$.

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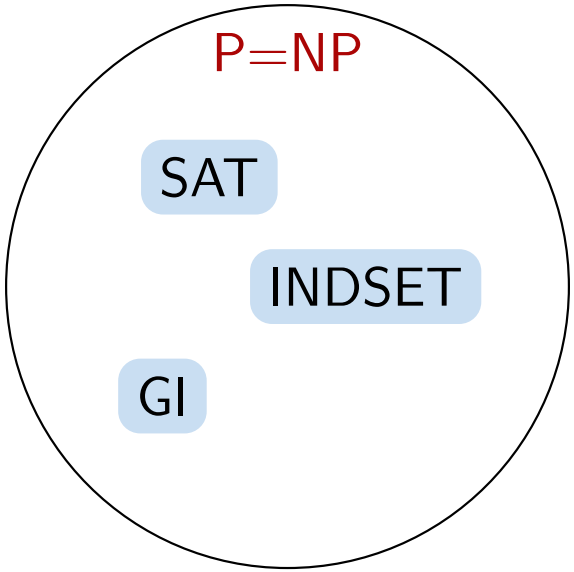

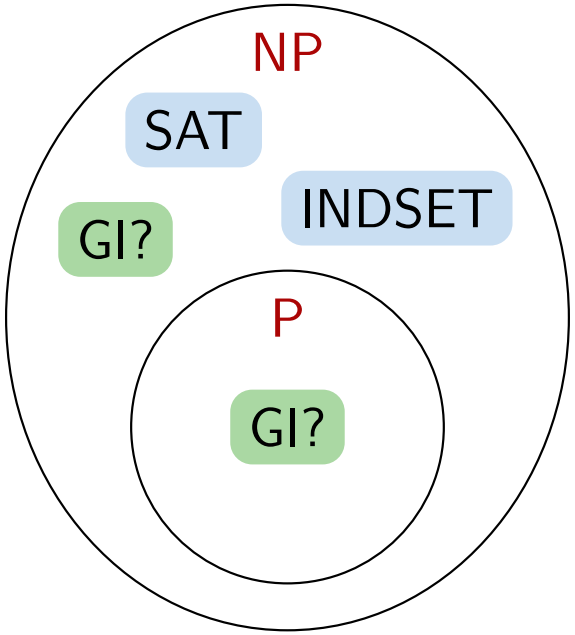

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(Also $INDSET \in P$ iff $P = NP$.)